Chinmay Nivsarkar

Email: chinmay.nivsarkar@nyu.edu | Phone: +1-551-344-7413 | LinkedIn: chinmay-nivsarkar

EXPERIENCE

ONEDOTSIX CONSULTING, Dallas, TX
Software Engineer - JP Morgan Chase Sep 2024 - Current

- Contributing to the data migration efforts across diverse on-premise and cloud applications following the acquisition of First Republic Bank by JP Morgan Chase, ensuring seamless integration and operational continuity.
- Developed and deployed Python-based automation tools to streamline migration processes, managing highly confidential data in a fast-paced agile environment while minimizing deployment errors and reducing manual effort by 60%.
- Orchestrated the reconciliation of tens of millions of documents across multiple systems and organizations, implementing gap analysis strategies and mitigation plans, resulting in improved data availability and compliance with regulatory standards.

SIEMENS HEALTHINEERS, Princeton, NJ
Research Intern - Large Language Models Jan 2024 - May 2024

- Implemented a proof-of-concept solution for extracting relevant cases from large datasets to generate subsets for training and evaluating Machine Learning and Deep Learning algorithms, using Semantic Search as the backbone.
- Created data ingestion and management workows for internal monitoring tools, improving turnaround times for existing processes by 25%.
- Contributed to the maintenance of legacy applications across various languages and tools and performed new feature enhancements.

SHELL INDIA MARKETS PVT. LTD, Bengaluru, India
Data Engineer Jul 2018 - Jul 2022

- Delivered $40M in cost savings and value realization by leading Azure migration and MVP deployments, overcoming technological limitations.
- Developed a custom Databricks framework (Scala, Spark SQL) for Azure migration, streamlined coding efforts for 99% of use cases.
- Spearheaded Azure platform upgrades, reduced refactoring needs by 80%, and crafted innovative Azure Data Factory pipelines.
- Led a team of 30+ developers in technical discussions, addressing challenges and designing solutions for upcoming use cases.
- Automated data workflows with a Python library, replacing Alteryx for SAP HANA CSV loads, saving 100+ hours monthly.

PROJECTS

Anomaly Detection in Network Traffic - Python, Tensorflow, sklearn Dec 2023

- Applied machine learning models to detect ARP spoofing attacks, enhancing network security by preventing man-in-the-middle attacks and safeguarding data integrity.
- Leveraged a comprehensive dataset featuring network traffic metrics (e.g., packet size, packet rate, inter-packet delays), analyzed through traditional ML models, achieving high accuracy Random Forest (99.99%) and GRU (99.76%).

Deepfake Detection - Python, PyTorch, NumPy, GANs May 2023

- Investigated various methods for detecting Deepfakes, leveraging GANs and comparing standard detection techniques with a novel approach utilizing a fine-tuned discriminator.
- Utilized the Flickr-Faces-HQ dataset to train a custom GAN for generating Deepfakes, then applied and compared different detection models, including CNNs and vision transformers.

Data Serving for Advanced Analytics - Azure Data Lake Storage, Databricks, Spark, Scala, MSSQL, PowerBI Jun 2022

- Leveraged newly available Spark 3.0 features and the Databricks SQL Endpoints to create a framework that would enable direct consumption from our Datalake into PowerBI while applying row-level security to the data.
- Engineered a scalable and cost-transparent solution for consumers, enabling data consumption use cases such as machine learning requiring large volumes (1M+ rows) that were not well supported by a reporting database.

No-Development Data Load Framework - Azure Data Factory, Databricks, Spark, Scala, MSSQL Apr 2021

- Extended the Databricks development patterns into a fully automated solution, enabling data loads using only configuration tables and a single Databricks notebook that could be seamlessly integrated into pre-existing Azure Data Factory pipelines.
- Reduced development effort to virtually zero for over 40% of use cases while supporting features such as custom joins, calculated columns, upserts, schema evolution, and more–saving 50-100 hours of effort per new deployment.

EDUCATION

NEW YORK UNIVERSITY, New York, NY
Master of Science in Computer Engineering May 2024

- Graduate Assistant for the High-Performance Machine Learning Operating and Systems courses.

MANIPAL INSTITUTE OF TECHNOLOGY, Manipal, India
Bachelor of Technology in Computers and Communication Engineering May 2018

SKILLS AND COURSEWORK

Languages: Python, Scala, Java, C#, C, Shell Scripting | **Databases:** MSSQL, SAP HANA, MongoDB, PostgreSQL
Others: Azure Data Factory, Databricks, Spark, NoSQL, Git, Power BI, Linux, High-Performance Computing, pandas, NumPy, Tensorflow, Pytorch, Algorithms, REST APIs, Docker, Kubernetes, Jules, Jenkins, Spinnaker, AWS
Coursework: Machine Learning, Deep Learning, High-Performance Machine Learning, Operating Systems, Data Engineering